BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Virtual Memory

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What happens to RAM and ROM contents when power is removed?

2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?

A) 1 B) 0
C) 2 D) 10

3. "A Boolean variable represents any unrestricted paragraph of text." Correct this. Explain why.

4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

Today's objectives

Explain when a computer needs virtual memory. Describe how the operating system moves pages between RAM and secondary storage.

Key vocabulary - Read, cover, recall

Virtual memory
Secondary storage used as backing space for RAM.

Page
Block of memory moved between physical and secondary storage.

Paging
Transferring memory pages between RAM and backing storage.

Thrashing
Severe slowdown caused by excessive page transfers.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

When RAM is insufficient	More working data may be required than physical RAM can hold at once.
Virtual memory	The operating system uses a reserved area of secondary storage as extra backing space for memory pages.
Moving pages	Pages are moved between RAM and secondary storage as needed; this is substantially slower than accessing RAM.
Performance consequence	Frequent movement between RAM and storage can cause a noticeable slowdown (thrashing).

I Do - One worked example

A laptop needs 6 GB for active programs but has only 4 GB available RAM. Explain what virtual memory does and why performance may fall. [4 marks]

1. Identify demand and capacity

2. 6 GB is needed but only 4 GB of physical RAM is available.

3. The OS can use some secondary storage as backing for inactive pages.

4. Accessing secondary storage is slower; swapping pages frequently slows the system. [4]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A laptop opens many browser tabs. Explain why virtual memory might be used. [3]

2. Explain why adding RAM could reduce use of virtual memory. [3]

3. A student says virtual memory makes a laptop faster than RAM. Evaluate the statement. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the output message when physical RAM is insufficient.

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
...
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State what type of storage is used to provide virtual memory. [1]

Q3. Explain two reasons why extensive paging can reduce performance. [4]

Q4. A program needs 5 GB but only 3 GB of free RAM is available. Explain the purpose of virtual memory in this scenario. [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An algorithm only gives reliable results when its inputs are handled as specified. ☐ T ☐ F
2. More working data may be required than physical RAM can hold at once. ☐ T ☐ F
3. The operating system uses a reserved area of secondary storage as extra backing space for memory pages. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 RAM loses its contents; ROM keeps its contents because it is non-volatile. ☐

Q2 A - 1 ☐

Q3 True or false. ☐

Q4 (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules. ☐

Code First - Model

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
print("May need virtual memory")
endif
```

Output: May need virtual memory

We Do - Check the model

1. The total working memory demand may exceed RAM [1]; the OS temporarily stores less-used pages on secondary storage [1] and brings them back when needed [1]. ☐

2. More active data fits in physical RAM [1], reducing page transfers to secondary storage [1] and therefore decreasing slow accesses [1]. ☐

3. Virtual memory extends apparent working capacity [1] and can allow more programs to remain open [1], but its secondary-storage backing is slower than RAM [1], so heavy paging reduces performance [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Secondary storage, e.g. SSD/HDD [1]. ☐

Q3 Pages transferred between RAM and storage [1], requiring additional input/output operations [1]; secondary storage is slower than RAM [1], so applications wait for required data [1]. ☐

Q4 Demand exceeds RAM by 2 GB [1]; some pages can be held in secondary storage [1] and restored to RAM when needed [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: